Nathan Palmer
Portfolio

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Work Experience

- **OpenSprinkler** *05/2024 - Present*
  Intern Amherst, MA
  - Implemented UI changes that added new interactive features as well as simplified and streamlined layouts.
  - Designed packaging for new products and their shipment.
  - Developed support for email notifications while improving other methods of notification such as IFTTT and MQTT.
  - Worked with weather APIs to improve and provide new features to the weather-controlled watering system.
  - Built, programmed, and tested Arduino-powered sprinkler controllers and other products to fulfill orders for thousands of units to customers around the world.

Education

- **University of Massachusetts Amherst** *Expected 05/2026*
  Bachelor's in Computer Science and Certificate in Design and Creative Technologies
  - GPA: 3.942/4.00

Skills

- **Creative Software:** Photoshop, Illustrator, InDesign, Premier, Figma
- **Programming Languages:** JavaScript, TypeScript, HTML, CSS, Python, C, C++, Java
- **Tools and Technologies:** Git, GitHub, Arduino

Extracurricular Activities

- **UMass Ballroom Dance Team** *09/2023 - Present*
  Secretary / Member
  - Worked as the secretary to record meeting notes, organize transportation for competitions, and register members in the competition tracking system.
  - Learned and practiced with the team for 5+ hours of practice every week to compete nationally with my dance partner.

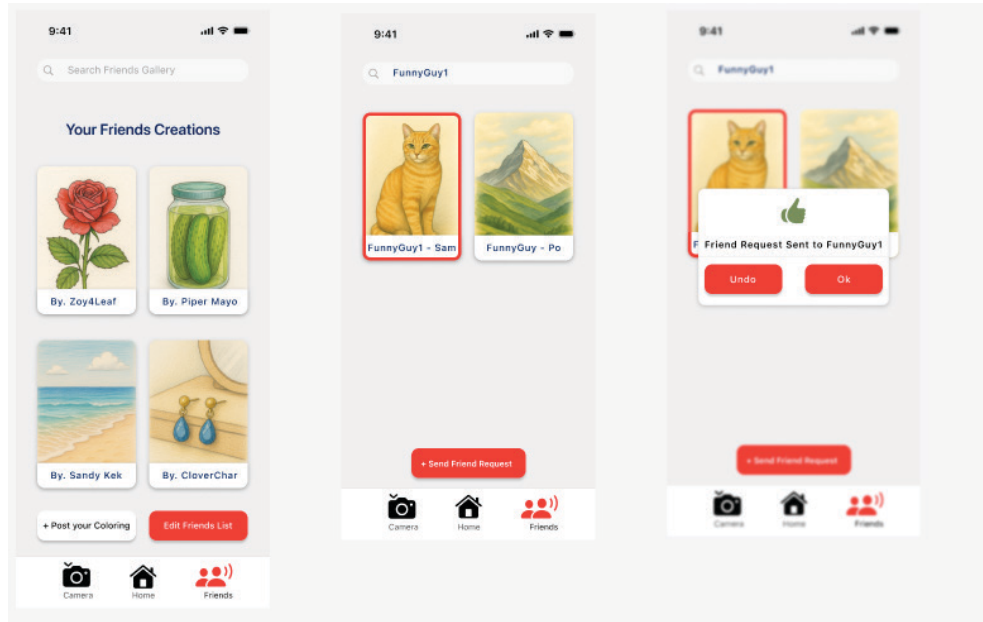
- **UMass Climbing Team** *09/2023 - Present*
  Fitness Instructor / Member
  - Practiced and prepared with the team at semiweekly practices to compete at the national level.
  - Led engaging and effective workout sessions alongside other instructors for groups of 30+ team members.

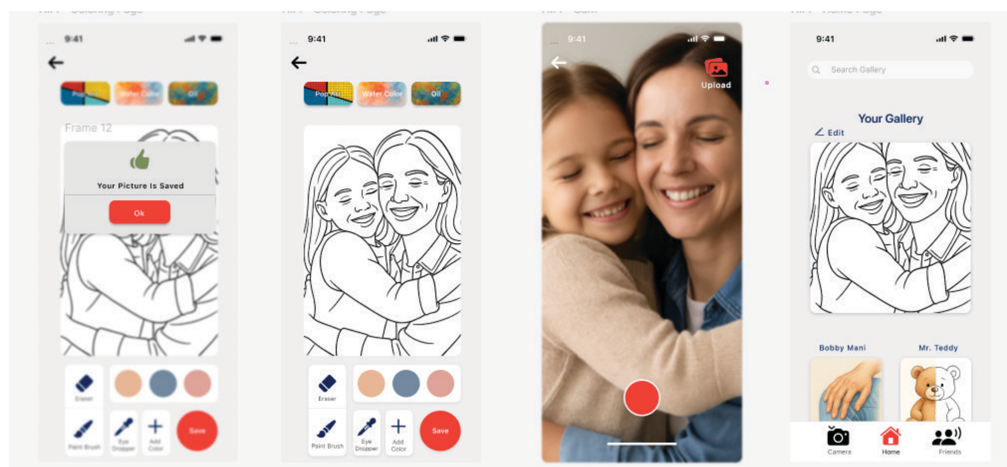
CustomColor

UX/UI Design Case Study (Full Pitch Doc Attatched)

09/25 - 12/25

Researcher / Designer / Collaborator

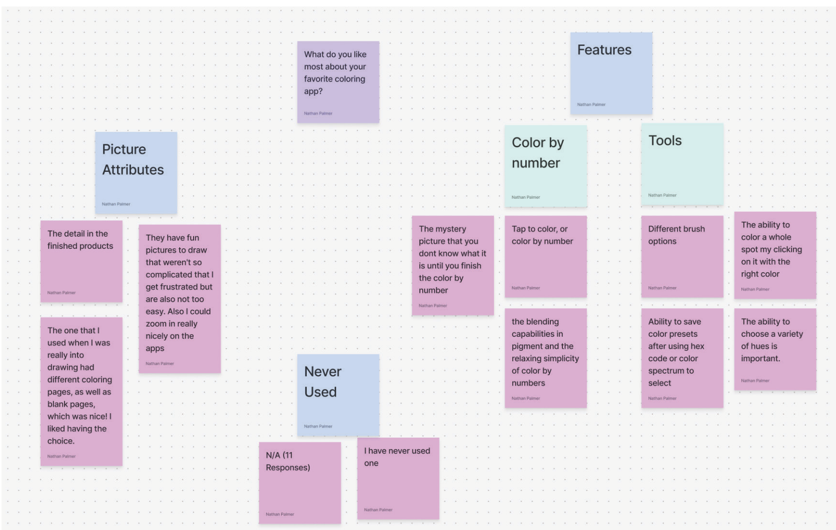




CustomColor is a case study done for an application that allows users to create and share custom coloring templates using images from their camera roll. The process culminated in a pitch document and final Figma prototype of the interface.

CustomColor

Methods



Data Collection

1. Google Form survey sent to family members, friends, affiliated sports teams
2. Multiple-choice, Multiple-answer (‘select all that apply’ checkboxes), Likert-scale, open-response questions for both quantitative and qualitative data collection
3. Bar charts and affinity diagrams for organization/ analysis.
4. Basic statistical analysis such as computing the mean/mode of survey results
5. Basic thematic analysis of qualitative results

Prototyping

Initial Prototype: Paper, Notability

Mid-Fi Prototype: Excalidraw

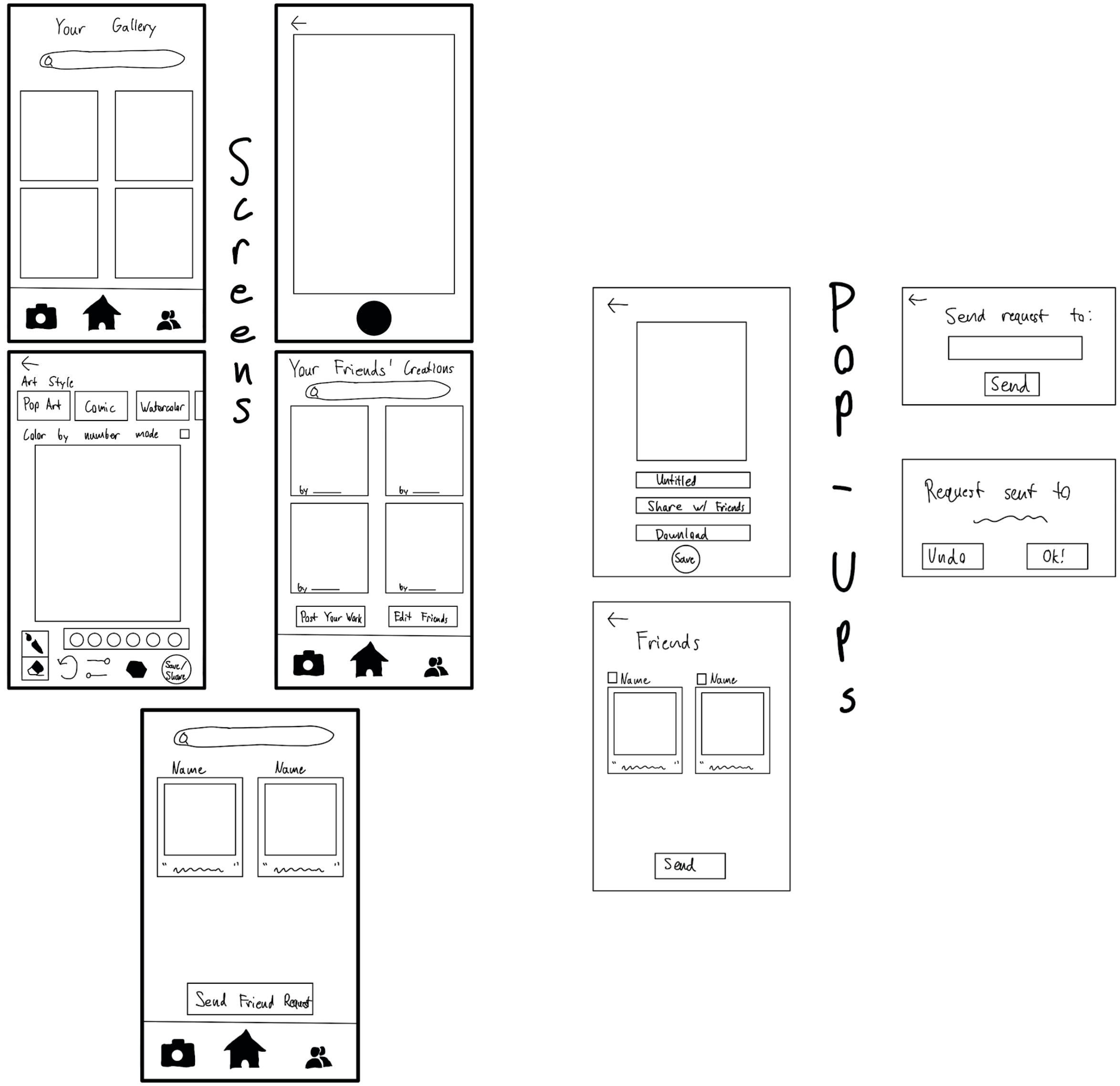
Revised and Final Prototype: Figma

Prototype Evaluation

1. Heuristic Evaluation
2. Usability Testing

CustomColor

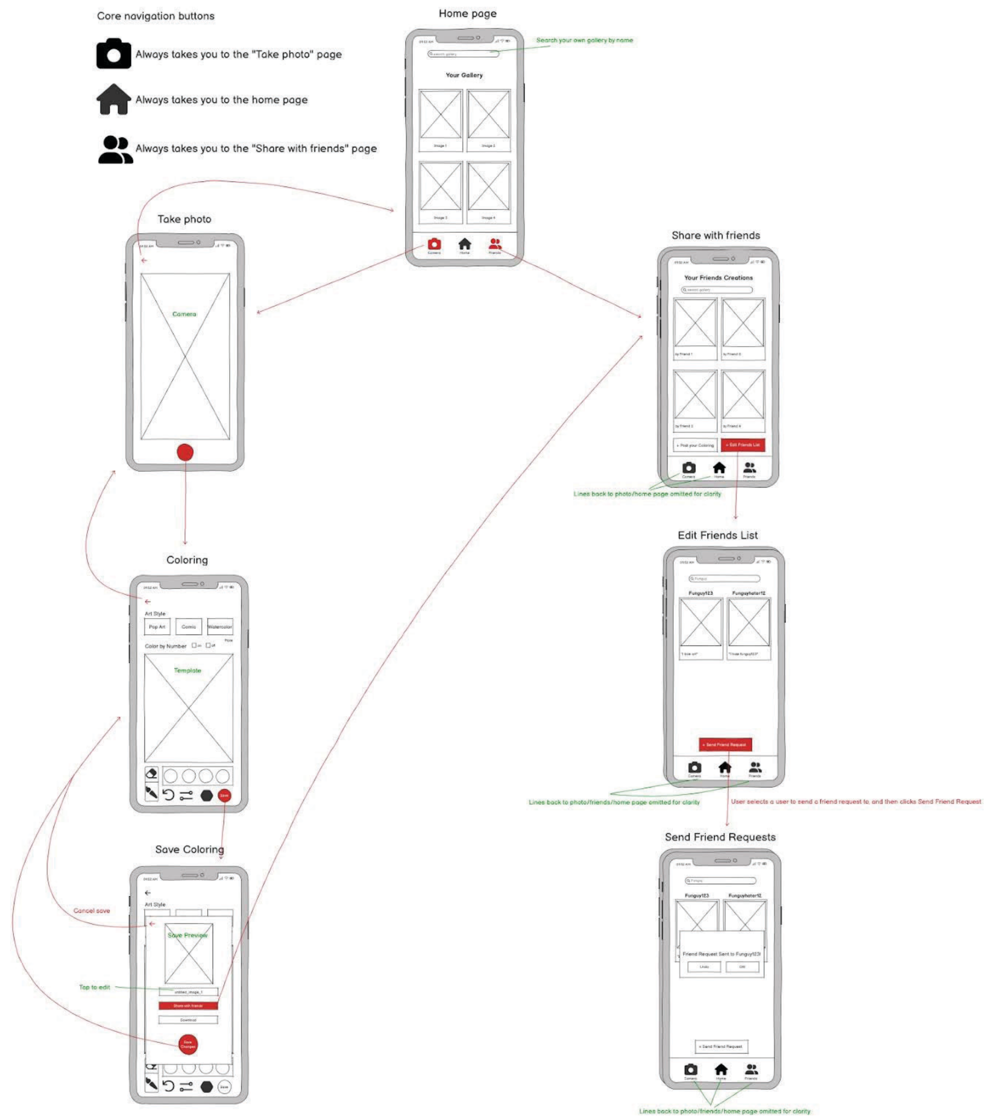
Lo-Fi Prototype



Our lo-fi prototype was based on our initial market research of current coloring apps and their reviews and survey answers. It was designed to be functional, with the intention to accomplish a list of tasks we prepared before beginning.

CustomColor

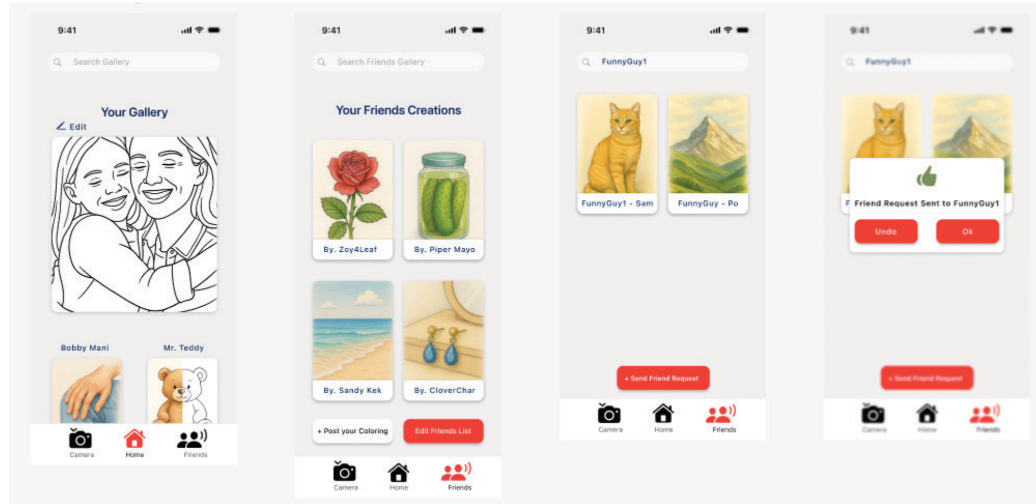
Medium-Fi Prototype

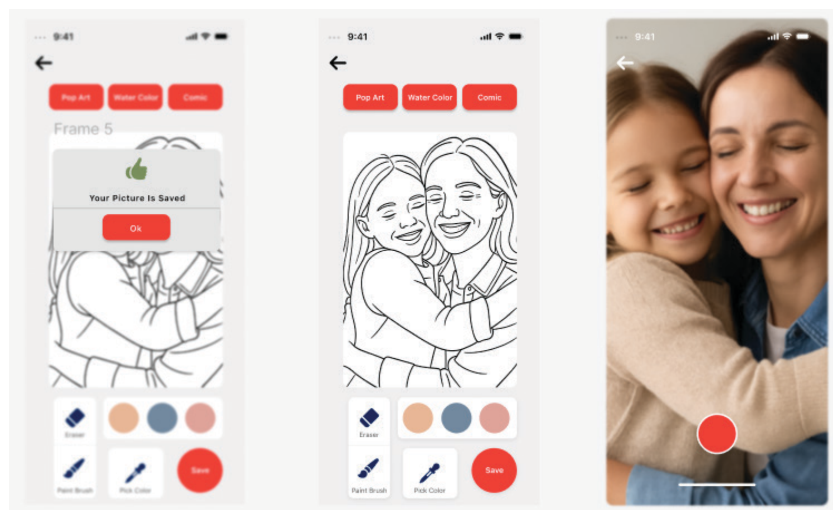


Moving to a higher fidelity stage, we began to look at mapping interaction, to ensure the application has an intuitive control flow. This helped us accomplish user testing in order to progress to the next stage in development.

CustomColor

Revised (Pre-Final) Prototype





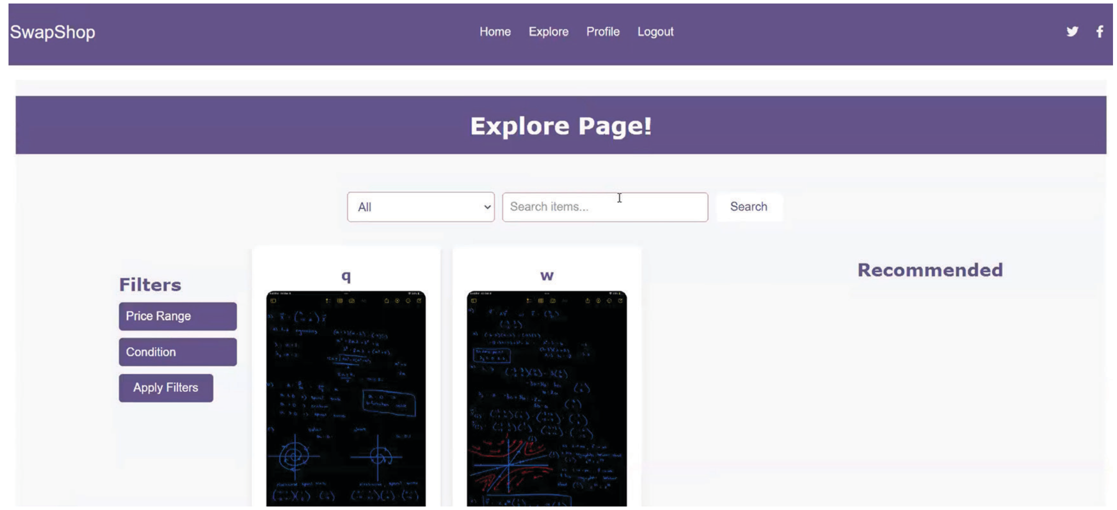
Our high-fidelity prototype was our second to last iteration. For this, we moved to Figma for a stronger design language, and based our prototype on user and team member evaluation and testing of the previous. This iteration served to polish up the visual identity of the app and prepare it for a final rround of testing.

SwapShop

Locally Hosted Website

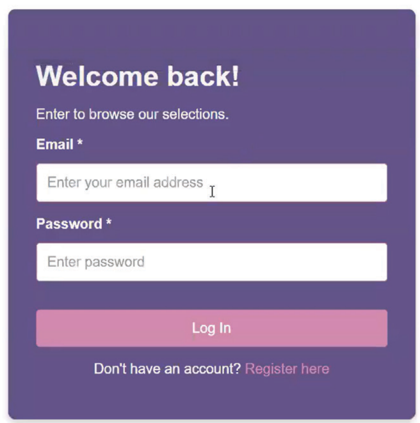
10/24 - 12/24

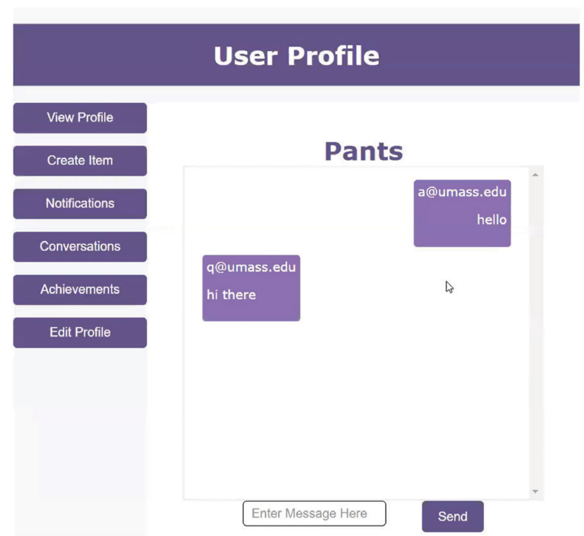
Designer / Developer /
Collaborator



SwapShop is a functioning mock-up of a buy/sell/trade platform for university students. As an extension of the local farmer's markets, it would allow students to buy, sell, and trade artwork, used clothing, and other items. It was created using HTML, JavaScript, and CSS.

The site site allows user registration, posting of products, curated and filtered search, and direct messaging via websocket.



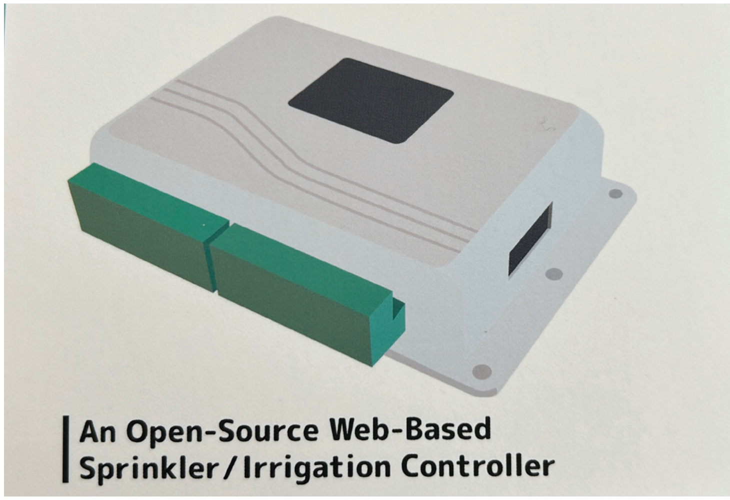


OpenSprinkler

Package Design

08/25

Designer



Illustrator

Indesign

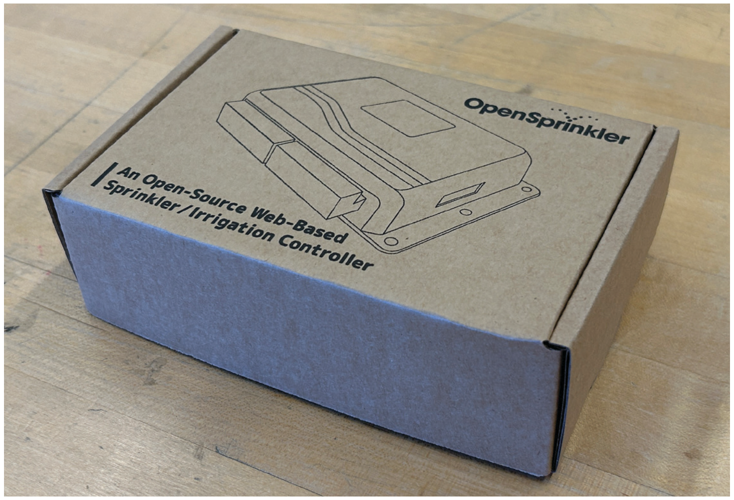
JLC Package Designer

OpenSprinkler

Exterior Package Design

08/25

Designer



Illustrator

Indesign

JLC Package Designer

This is Water

Booklet Design

04/25 - 05/25

Designer

True, there are plenty of religious people who seem arrogant and certain of their own interpretations, too. They're probably even more repulsive than atheists, at least to most of us. But religious dogmatists' problem is exactly the same as the story's unbeliever: blind certainty, a close-mindedness that amounts to an imprisonment so total that the prisoner doesn't even know he's locked up.

The point here is that I think this is one part of what teaching me how to think is really supposed to mean. To be just a little less arrogant. To have just a little critical awareness about myself and my certainties. Because a huge percentage of the stuff that I tend to be automatically certain of is, it turns out, totally wrong and deluded. I have learned this the hard way, as I predict you graduates will, too.

Here is just one example of the total wrongness of something I tend to be automatically sure of: everything in my own immediate experience supports my deep belief that I am the absolute centre of the universe; the realest, most vivid and important person in existence. We rarely think about this sort of natural, basic self-centredness because it's so socially repulsive. But it's pretty much the same for all of us. It is our default setting, hard-wired into our boards at birth. Think about it: there is no experience you have had that you are not the absolute centre of.

The world as you experience it is there

in front of YOU

or behind YOU,

to the left or right of YOU,

on YOUR TV

or YOUR monitor.

And so on.
Other people's thoughts and feelings have to be communicated to you somehow, but your own are so immediate, urgent, real.

Procreate

Indesign

CustomColor

Jason Alexander
Breanna Gushman
Colin Leary
Nathan Palmer

So, what is CustomColor?

Great question! CustomColor is a coloring application meant to bring the things that mean the most to you to life! We do not want to leave users to preset designs, we want to inspire personal creativity, which is why CustomColor allows users to take their own photos and transform them into coloring sheets. Then, for those who want a simple coloring experience with some structure, a color by number mode is ready for them. Meanwhile, those looking for a little more freedom can work with a variety of brushes and colors in free coloring mode. Then, to let users share their creativity, these templates and finished coloring pages can be shared to friends within the app or even to the public templates list for anyone to try!

Why is CustomColor so great?

People who enjoy coloring are limited to pre-made designs and cannot easily personalize their experience. Even if that option is provided, coloring applications are distinctly divided into two separate categories of color by number and free color. CustomColor grants people the freedom to color what inspires them the most, leading to a very fulfilling, meaningful, and enjoyable experience. Art and creativity are essential in our innovation focused modern world, and we are here to merge the benefits of modern technology with artistic freedom, encouraging an imaginative form of self-expression and relaxation. We want to provide a one stop shop for people looking to relax, create, and share.

Why should I choose custom color?

When looking for an application to fill this creative niche right now, there are always sacrifices that have to be made given the current options. When given the option to create your own template, image conversion can get sloppy and lead to overcomplicated, dark, and hard to color templates. When moving into other aspects of customization, some apps, especially color by number apps, have small settings menus that do not give users the freedom to customize their experience to suit their needs. This lack of customization combined with frequent advertisements lends to frustration from users in an activity meant for relaxation. On top of that, various apps on the app store have issues with user experience such as inconsistent brush settings, difficult blending, or lack of tutorial that we have researched and pushed to avoid. We want a seamless experience in order to provide a creative outlet that will never hinder the user.

Data Collection

Motivations/Research Questions

1. Do users already use similar apps? If so, which ones, and why?
2. For those who have not used apps like this before, what app features/characteristics would inspire them to use our app?
3. How frequently and for how long do users engage in coloring or similar activities?

Methods Employed

1. Google Form survey sent to family members, friends, affiliated sports teams
2. Multiple-choice, Multiple-answer (‘select all that apply’ checkboxes), Likert-scale, open-response questions for both quantitative and qualitative data collection
3. Bar charts and affinity diagrams for organization/analysis.
4. Basic statistical analysis such as computing the mean/mode of survey results
5. Basic thematic analysis of qualitative results

Data Collection

Finding	Methodology	Impact
For the group surveyed, phones, laptops, and tablets (in that order) were the most used devices for creativity and relaxation.	This data was collected via our Google Form survey as a multiple-answer checkbox question. The data was presented as a bar chart and analyzed quantitatively (which bars are numerically largest?)	Validated the idea that a digital coloring app could be accessible and used by a large number of people. It also suggested that our UI designs should focus on phones, though other screen sizes were also worth considering.
The vast majority of respondents rarely or never use coloring apps.	This data was collected via our Google Form survey as a multiple-answer checkbox question. The data was presented as a bar chart and analyzed quantitatively (which bars are numerically largest?)	This encouraged us to make sure that our app would be memorable and easy to return to after a long period of inactivity. The user interface should be welcoming and intuitive for people trying out coloring apps for the first time.
The most common reasons respondents listed for coloring were relaxation and stress relief.	This data was collected via our Google Form survey as a multiple-answer checkbox question. The data was presented as a bar chart and analyzed quantitatively (which bars are numerically largest?)	This encouraged us to make sure our app invokes peacefulness/serenity.
Many respondents cited complexity and advertisements as two aspects of coloring apps they dislike.	This data was collected via our Google Form survey as an open response question. The data was organized into an affinity diagram and analyzed thematically (what important patterns/ideas can be identified in the responses?)	Through our prototypes we kept working on making the user experience more intuitive and simple to lower complexity. We also ensured that the app would not have any advertisements to not bother users.

Data Collection

Finding	Methodology	Impact
Most respondents cited specific features as their most liked aspects of their favorite coloring apps. These responses are roughly divided by color-by-number features and special drawing tools (such as the ability to choose a variety of hues)	This data was collected via our Google Form survey as an open response question. The data was organized into an affinity diagram and analyzed thematically (what important patterns/ideas can be identified in the responses?)	Motivates that color-by-number in particular would be a good central feature of our app. Additional features, like a custom color selector would also help make the app more enjoyable for users.
The top three most important features according to the respondents (in order) are a wide color palette, realistic brush tools, and no ads/paywalls.	This data was collected via our Google Form survey as a series of multiple choice selections, where users selected a numerical score (1-5) for the importance of a proposed feature. The data was presented as a set of bar charts and analyzed quantitatively (which bars are numerically largest, and what is the mean ranking of importance for each proposed feature?)	We ensured that our prototypes would include a color palette customization feature, realistic brush tools, and no ads/paywalls.
The majority of respondents were at least partially interested in uploading their own photos to turn into coloring pages. (specifically, the mode was ‘yes’, with the next most common being ‘maybe’)	This data was collected via our Google Form survey as a multiple-choice question. The data was presented as a bar chart and analyzed quantitatively (which bars are numerically largest?)	This encouraged us to explore the idea of having users upload their own photos and automatically turn them into coloring-by-number templates.

Data Collection

Reflection

Most of our collected data supported our initial hypotheses of what a successful color-by-number application should look like. Specifically, the results encouraged a clean, ad-free, accessible mobile app, with specific features like color customization and uploading your own photos.
What went well was that we did not have much trouble getting an adequate number of respondents, and the information we got led to some important guiding information (for example, the app should be designed to welcome users who are new to coloring book apps, as many people seem to not use them much).
More emphasis could have been placed on investigating how users would respond to a sharing/friends feature, as this was a core component of our future prototypes.

Initial Prototype

Design Decision	Motivation	Relevent Usability Goals/Design Concepts
Homepage shows users' own colorings in a card grid	Helps users immediately see what the main purpose of the application is. This is because the data collection highlighted that respondents valued an intuitive/simple experience.	Visibility, Recognition over recall, Card-based grouping
Search bar placed at the top of Friends/Gallery pages	Provides a fast way to find specific people or images, especially if they have many friends. Top placement follows standard mobile conventions.	Mapping, Conventions, Efficiency, Mental models, Learnability, Fitts' Law (easy-to-access location)
Confirmation popup with thumbs-up icon ("Friend Request Sent") and two buttons: Undo + OK	Reinforces success with a visual signifier and gives the chance for users to undo a friend request, which might otherwise create awkward interactions.	Feedback, Error recovery, Affordances, User control & freedom, Reversibility
Painting page uses bottom toolbar (Eraser, Paint Brush, Eye Dropper, Add Color, Save/Share)	Groups editing actions together and follows industry standards for drawing apps. Helps users quickly identify tool functions without reading text. Save/Share was combined to save space and because they are conceptually similar actions.	Grouping (Gestalt), Consistency, Learnability, Signifiers, Aesthetic usability
Save confirmation modal	Reinforces success with a visual signifier, reducing the chance the app is inadvertently creating stress rather than reducing it.	Positive feedback, Visibility, User confidence

Initial Prototype

Design Decision	Motivation	Relevent Usability Goals/Design Concepts
Camera upload screen with centered shutter button	Mirrors the layout of standard photo apps, making the interaction effortless. Directly supports user workflow of capturing a photo → converting to template.	Mental models, Mapping, Consistency with real-world apps, Discoverability
Main navigation tabs at the bottom (Camera, Home, Friends)	Helps users jump between the 3 core flows without getting lost. This style matched apps our group members were familiar with (Snapchat, Clash Royale). Our hope was that this layout was also familiar to users.	Navigation mapping, Consistency, Learnability

Evaluation of Initial Prototype

Finding	Methodology	Relevant UI Elements	Impact
Color palette switcher is confusing	Post-task questions as well as observation during testing	Color Palette switcher	Changed palette switcher to a drop down menu, more specific names
There is a weird stop sign shape button that doesn't visibly do anything (it was a color switcher but difficult to tell in black and white)	Post-task questions	Color Picker	Changed color picker to an array of buttons rather than one single menu
Difficult to tell what buttons do what when they are in black and white	Post-task questions	All color-related buttons	Added color to Hi-Fi prototype
Save, export, and addle titles buttons are confusing	Watching participants accomplish the tasks	Save button, Export button, add title box	Changed shape and size of these buttons so they are not all identical.

The initial prototype and subsequent evaluation was a somewhat rocky stage in this project. During the early stages of the semester, we were pretty unaware of what a Lo-Fi prototype actually meant, so we accidentally created a medium-fidelity prototype in its place. We then had to go back and remake a less detailed prototype later. As for how the evaluation went, I would say overall very strong. Our test subjects were able to navigate the UI quickly and easily up until the task which asked them to change the color palette they were using. Most subjects voiced this step was confusing as to what the color palette actually was, let alone how to change it. We attributed this problem to the Lo-Fi prototype being in black and white, however it continued to plague us for the rest of the project.

Methods Used

1. Usability Testing
2. Heuristic Testing

Revised Prototype

Design Decision	Motivation	Relevent Usability Goals/ Design Concepts
Added colors	Due to the nature of our app, the lack of color in the lo-fi prototype was a significant detraction from overall usability	Signifier: colored buttons easily convey that they changed the color of the brush, whereas a non-colored button is merely a circle
Added mock camera footage	Participants weren't aware when the "camera" was on	Signifier: shows users the "camera" is functional by showing the readout on screen
Added Eyedropper option	During both of our usability testing users were confused on how to add their own color. First design we used a hexagon which was very unclear. So we switched it to an eye dropper and labeled it "pick color".	Nielsen Heuristic: Prevent errors, Visibility of required action
Cleaned up UI	Participants voiced, along with group members, that the UI we had in the template screen was too messy. In order to remedy this, we removed outlines from buttons (needless) and moved some elements to the top of the page and out of the way rather than on the bottom, where they clutter more important buttons	Simplicity: Design is easier to interpret after the changes. Thus, it is also quicker to learn and easier to remember once learned.

Evaluation of Revised Prototype

Finding	Methodology	Relevant UI Elements	Impact
Edit Friends list is confusing	Post-task interview	Friends list tab	The phrasing of "Edit Friends list" was confusing when the task was to add a friend. This finding is corroborated by other popular websites, who generally only use "Edit" to mean delete in terms of friends. We changed the phrasing to "Add friends" and moved delete functionality elsewhere
Hexagon & color picker to select color is confusing	Post-task interview	Change from Hexagon → eyedropper button	By dividing tasks with buttons, the UI is clearer to users. Verses a hexagon which could mean anything.
Difficult to tell what buttons do what when they are in black and white	Post-task questions	All color-related buttons	Added color to Hi-Fi prototype
Save, export, and addle titles buttons are confusing	Watching participants accomplish the tasks	Save button, Export button, add title box	Changed shape and size of these buttons so they are not all identical.

Methods Used

1. Usability Testing

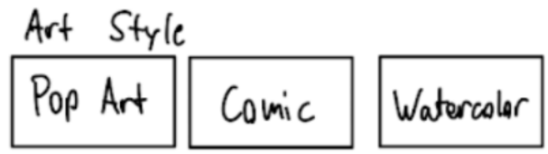
Evaluation of Revised Prototype

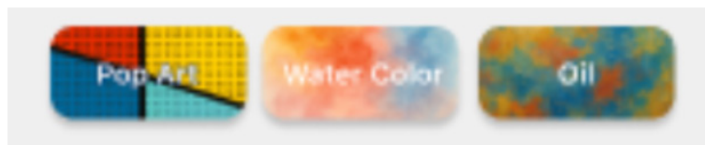
The revised prototype was a very strong stage of this project for our group. Having ironed out some color palette picker issues from the Lo-Fi prototype and having the luxury of adding color this time, our app's usability saw significant improvement. Participants voiced satisfaction with how the app felt, as though it felt more like a product than a prototype. We discovered a slight lack of popups, specifically when adding friends. Users expressed they wanted a "Friend added!" screen to see if their friend request went through or not. Additionally, when users take a photo, the app instantly turns the photo into a template. While not mentioned by the participants, the observers noted this could lead to extreme annoyance were the photo taken to be subpar or even entirely accidental. Since there was no photo confirmation screen, processing power and user's time would be wasted turning the useless photo into a template before they could back out. In order to remedy this, we added "Confirm" and "Retake" buttons to the final prototype. Overall, the three participants who volunteered to help test our Hi-Fi prototype gave us some very valuable feedback that we brought to the final model.

Final Prototype

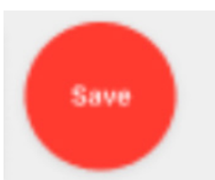
How well did we meet the various Usability Goals?

I believe that we did a good job on improving learnability/memorability over our prototype iterations. Specifically, it was originally quite hard to distinguish between changing color palette and changing template style. By changing the template style selection buttons to visually look like the style changes, a clearer mental distinction between template style and color palette can be made. See the two figures below to see the comparison.





Similarly, learnability improved with iterations with respect to the save/share feature. Initially, these two features were accessed via a combined button, which was unconventional and overcomplicated. Our future prototypes simplified this design and added color coding to make the save feature more visible, aiding learnability. See the two figures below.



Other usability goals were successfully accomplished, such as error prevention through eraser tools, confirmation modals for potentially harmful situations (sending a friend request, for example).

Usability was generally achieved, as we included in our prototype the main features desired by our initial data collection process, though certain features that respondents wanted, such as a timelapse of one's coloring, were omitted for simplicity. In future prototypes, we could add a few additional features, though this would have to balance the number of features with learnability and memorability.

Final Prototype

The presence of various design concepts

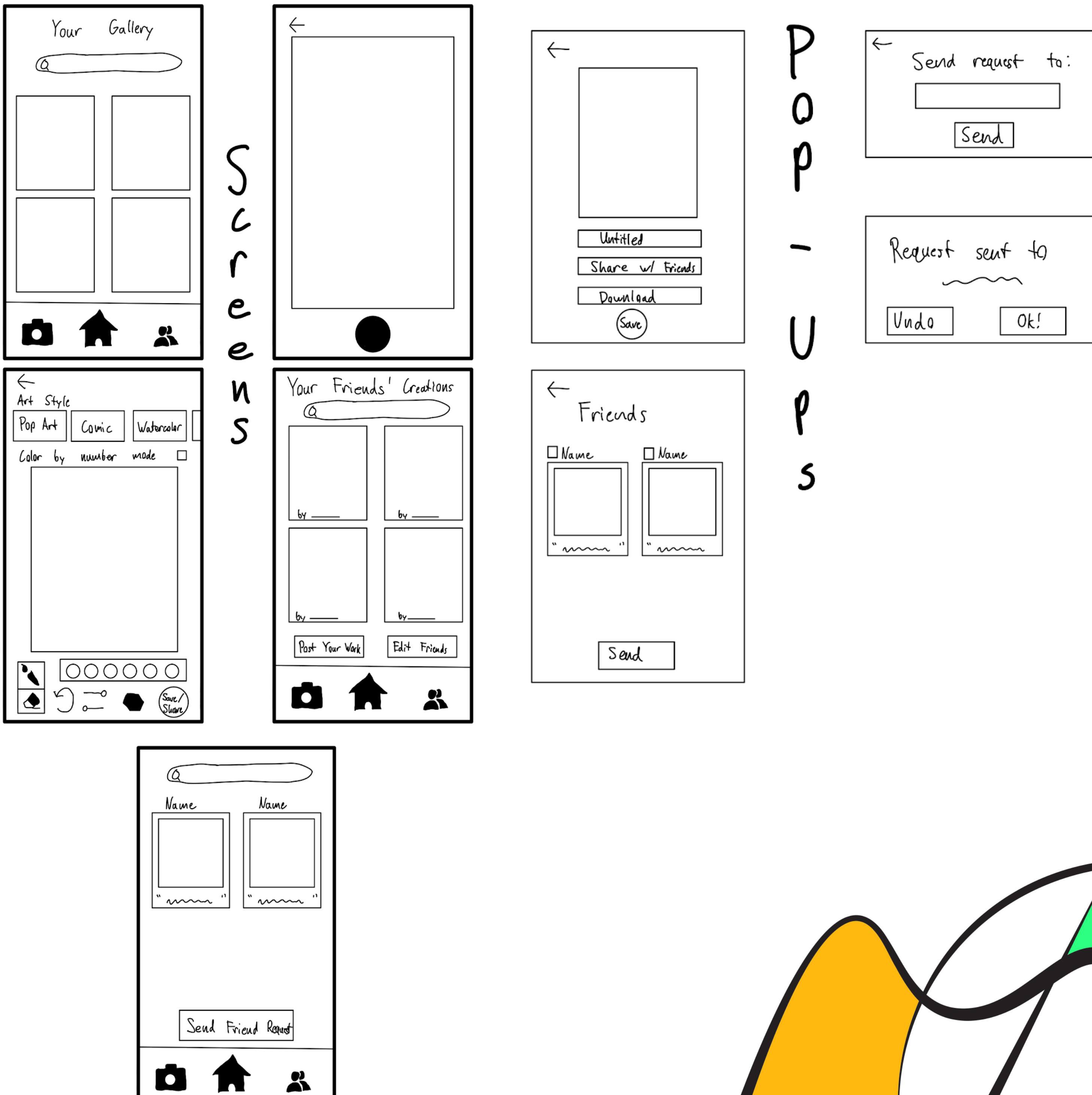
I think that we effectively use modals to show feedback, such as when you send a friend request or save your work. We also try to utilize the transfer effect whenever possible, such as the placement of the search bar, the camera modal, the bottom three navigation buttons, the pencil/eraser icons for template filling.

Motivation for the use of specific interaction types

We focus on instructing and manipulating interaction types, as our app uses menus for users to navigate features, and instructing is the simplest way to achieve this. Manipulation is used for the coloring book features, as we attempt to replicate the feeling of filling in a physical color-by-number page.

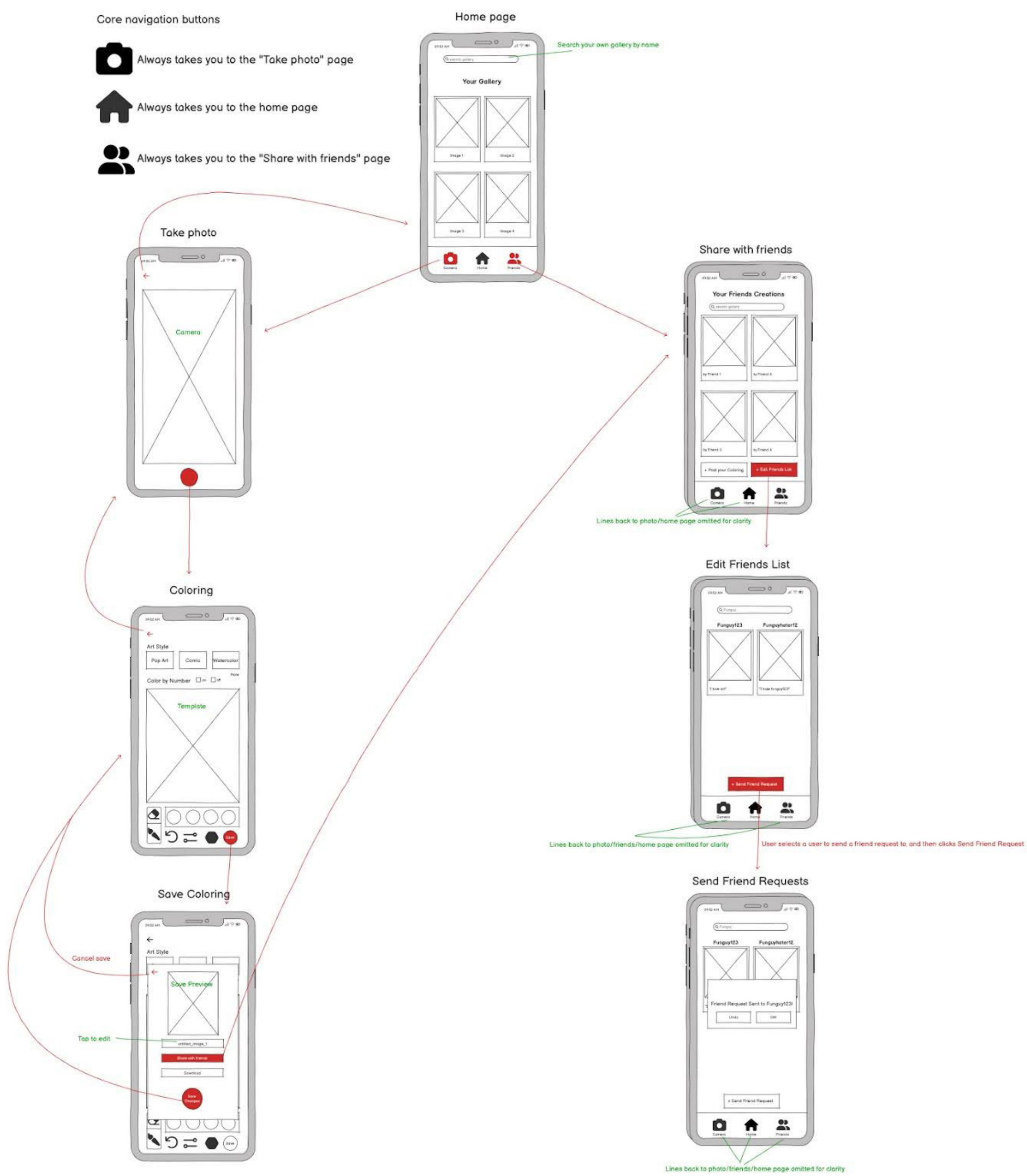
Prototypes

Initial Prototype



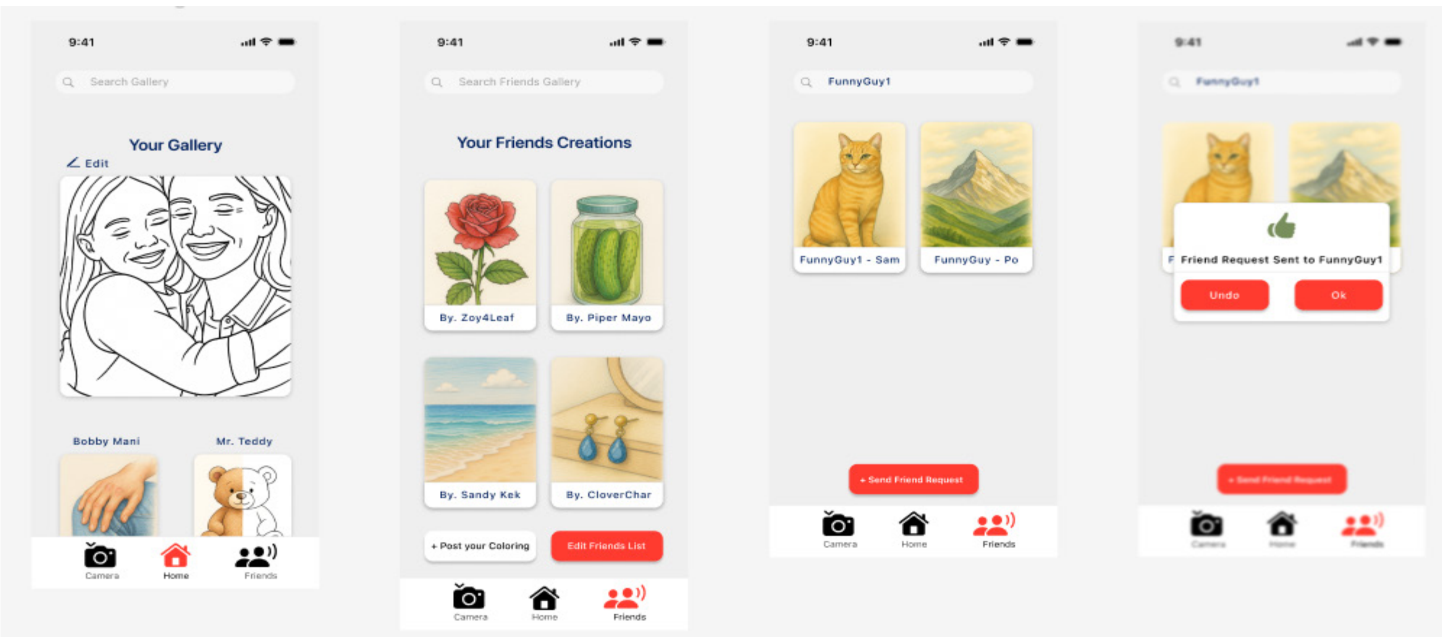
Prototypes

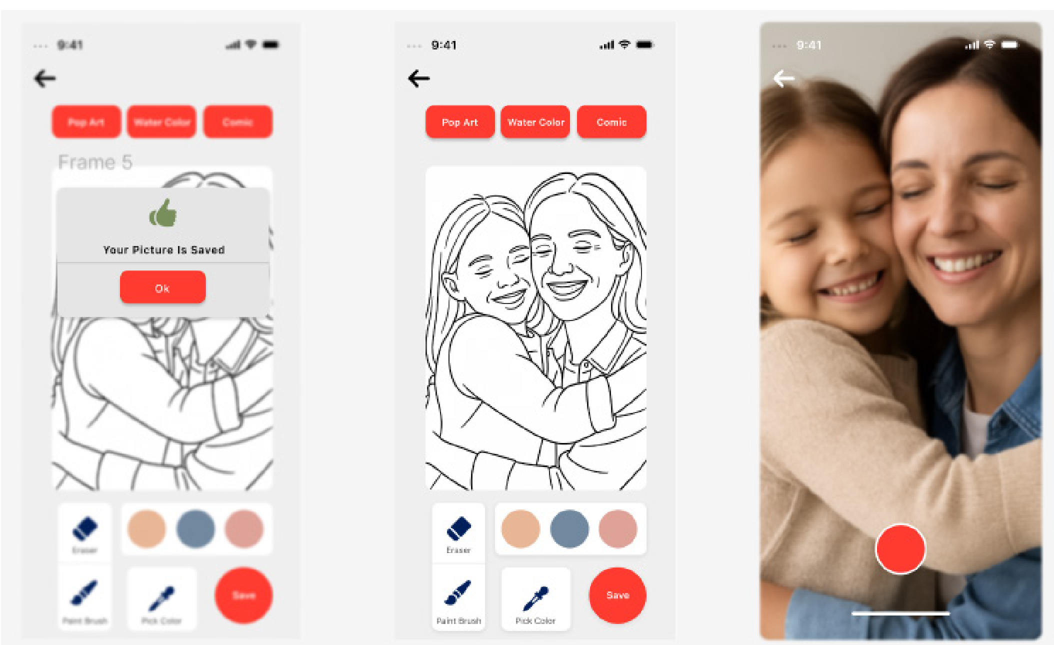
Initial (Medium Fidelity) Prototype



Prototypes

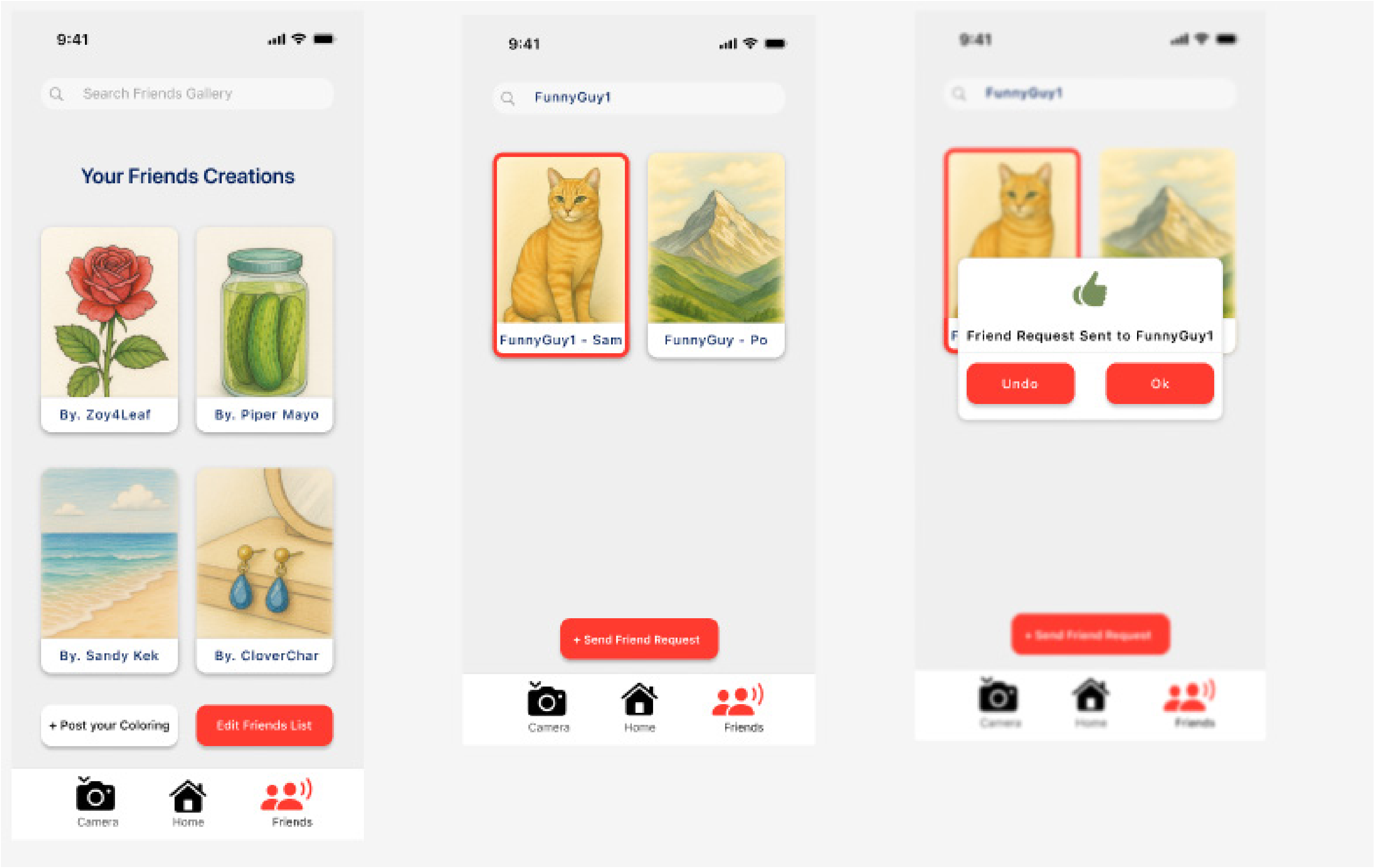
Revised Prototype





Prototypes

Final Prototype



Prototypes

Final Prototype